Research Topics in Blockchain
&
Blockchain and Its Application

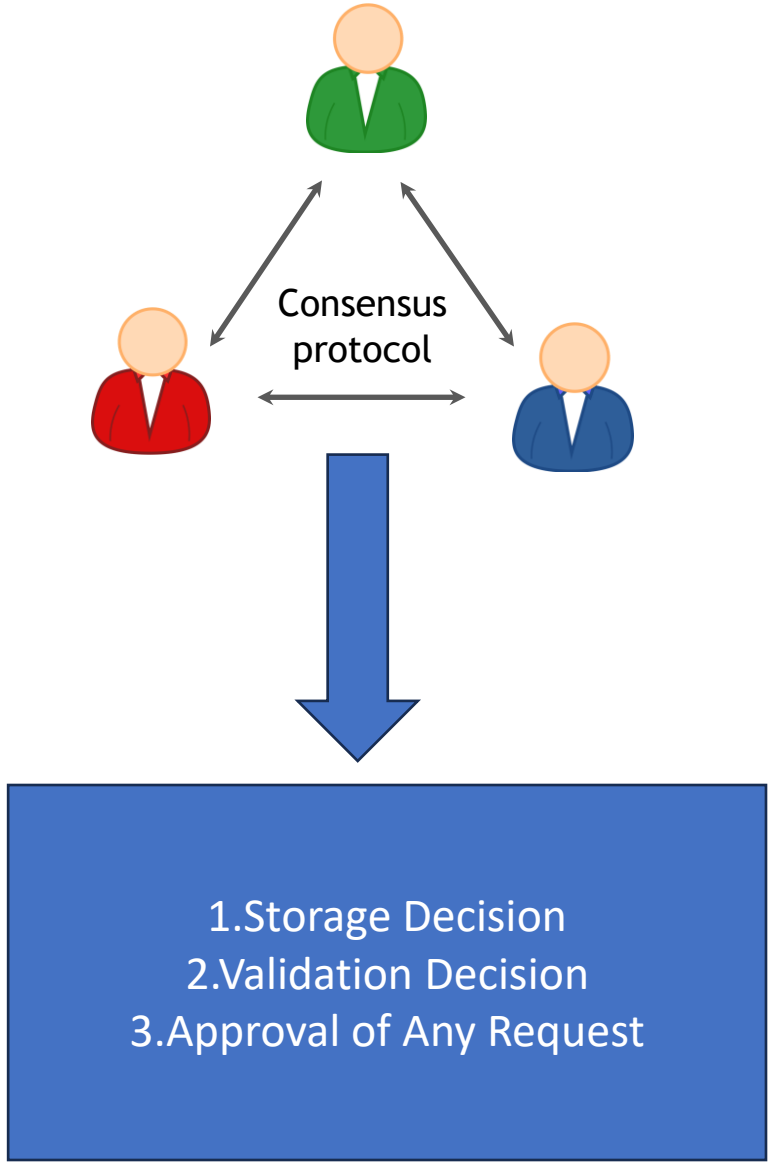


Lecture # 8:
Decentralization and Consensus Mechanism

- ✓ **Decentralization**
- ✓ **Distributed Decentralization**
- **Lab 2 Demonstration**

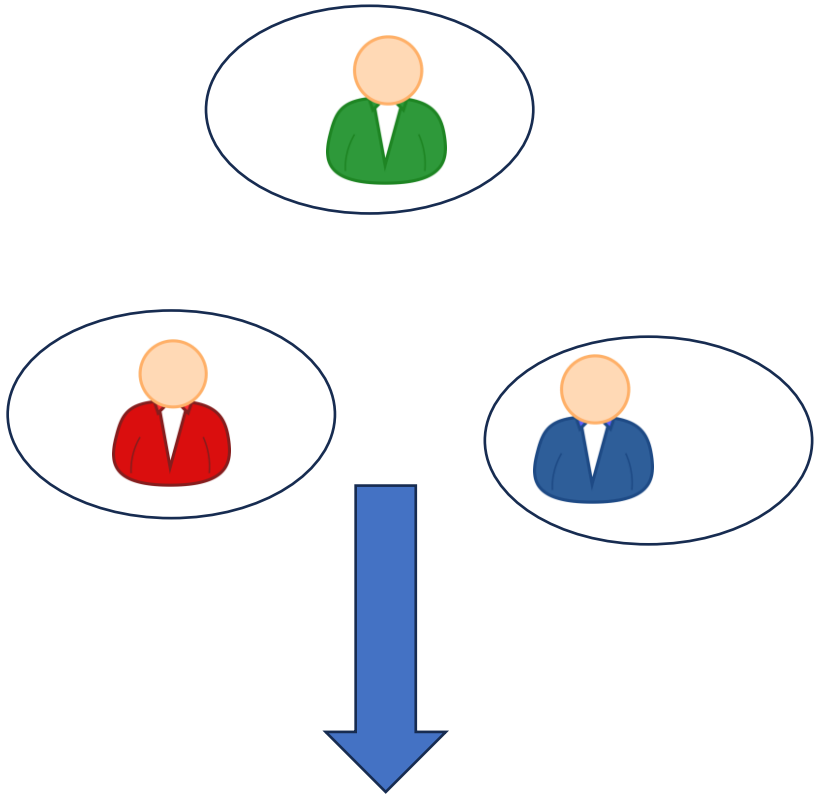
Distributed Decentralization

- ✓ Involves multiple independent nodes with no central authority
- ✓ Governed by consensus mechanisms. This structure emphasizes both distribution and decentralization, enhancing security and redundancy through distributed data storage.
- ✓ Examples include blockchain networks like Bitcoin and Ethereum, as well as decentralized file systems like IPFS and BitTorrent.

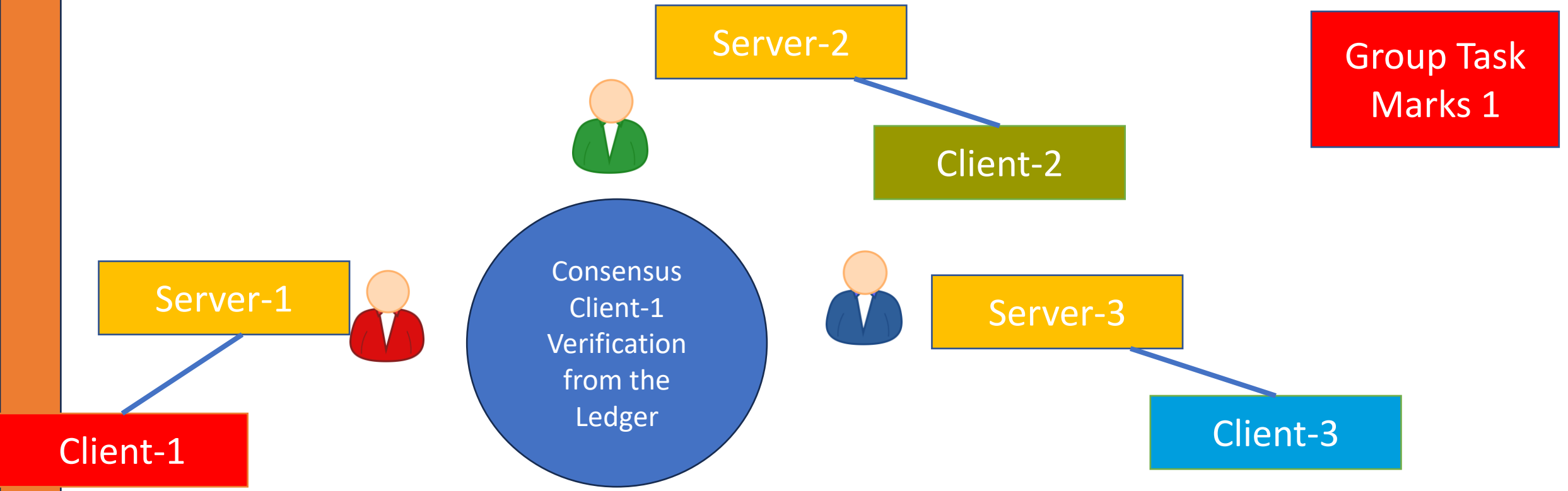


Decentralization

- ✓ Refers to the spreading out of control across multiple entities, though it may still retain some central oversight.
- ✓ It's less about independent nodes and more about shifting authority from a central body to a distributed set of entities.
- ✓ For instance, in multi-cloud infrastructure or federated governance models, data and control are distributed, but coordination might still be centralized in some instances.



Implementation Distributed Decentralized Network For Consensus Mechanism For Recording Hello Message in Ledger



Implementation Distributed Decentralized Network For Consensus Mechanism For Recording Hello Message in Ledger

How can we implement a **distributed decentralized network** for a **consensus mechanism solution**, where three servers, each having their own clients, share a single **JSON-based ledger** containing client identity hashes for consensus?

Specifically, one of the servers is deliberately selected to store transactions and validate client hash requests by verifying them from the ledger.

For example, suppose **Client 1 from Server 1** sends a "Hello" message. How should the consensus mechanism handle the verification and validation of this transaction across the network?
What are the best practices and potential challenges in designing such a system to ensure security, integrity, and fault tolerance?

Implementation Distributed Decentralized Network For Consensus Mechanism Solution

